Database System Engineering and Distributed Backend Development

Course Code: 25CS13002E

ROLL NO:2520030414

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Query 1: -CREATE DATABASE

```
CREATE DATABASE IF NOT EXISTS bookflow0_db;
USE bookflow0_db;
```

Output: -

12	17:07:56	CREATE DATABASE IF NOT EXISTS bookflow0_db	1 row(s) affected
13	17:08:04	USE bookflow0_db	0 row(s) affected

Query 2: -CREATE TABLE

```
CREATE TABLE orders (
    ord_no INT PRIMARY KEY,
    purch_amt DECIMAL(10,2),
    ord_date DATE,
    customer_id INT,
    salesman_id INT
);
```

Output: -

14	21:36:23	CREATE TABLE orders (	ord_no INT PRIMAR...	0 row(s) affected	0.125 sec

Query 3: - INSERT

```
INSERT INTO orders VALUES
(70001,150.50,'2012-10-05',3005,5002),
(70009,270.65,'2012-09-10',3001,5005),
(70002,65.26,'2012-10-05',3002,5001),
(70004,110.50,'2012-08-17',3009,5003),
(70007,948.50,'2012-09-10',3005,5002),
(70005,2400.60,'2012-07-27',3007,5001),
(70008,5760.00,'2012-09-10',3002,5001),
(70010,1983.43,'2012-10-10',3004,5006),
(70003,2480.40,'2012-10-10',3009,5003),
(70012,250.45,'2012-06-27',3008,5002),
(70011,75.29,'2012-08-17',3003,5007),
(70013,3045.60,'2012-04-25',3002,5001);
```

Output: -

15 21:37:55 INSERT INTO orders VALUES (70001,150.50,'20... 12 row(s) affected Records: 12 Duplicates: 0 Wa... 0.047 sec

Query 4: - WHERE CLAUSE

```
SELECT *
FROM orders
WHERE purch_amt > 2000;
```

Output: -

16 21:40:16 SELECT * FROM orders WHERE purch_amt > 20... 4 row(s) returned 0.000 sec / 0.000 sec

Result Grid Filter Rows: Edit:

ord_no	purch_amt	ord_date	customer_id	salesman_id
70003	2480.40	2012-10-10	3009	5003
70005	2400.60	2012-07-27	3007	5001
70008	5760.00	2012-09-10	3002	5001
70013	3045.60	2012-04-25	3002	5001
NULL	NULL	NULL	NULL	NULL

Query 5: -WHERE CLAUSE

```
SELECT *
FROM orders
WHERE ord_date='2012-09-10';
```

Output: -

17 21:42:11 SELECT * FROM orders WHERE ord_date='2012... 3 row(s) returned 0.031 sec / 0.000 sec

Result Grid Filter Rows: Edit:

ord_no	purch_amt	ord_date	customer_id	salesman_id
70007	948.50	2012-09-10	3005	5002
70008	5760.00	2012-09-10	3002	5001
70009	270.65	2012-09-10	3001	5005
NULL	NULL	NULL	NULL	NULL

Query 5: -ORDER BY

```
SELECT *
FROM orders
ORDER BY purch_amt DESC;
```

Output: -

18 21:43:46 SELECT * FROM orders ORDER BY purch_amt ... 12 row(s) returned 0.000 sec / 0.000 sec

Result Grid | Filter Rows: | Edit: | Export/Im

ord_no	purch_amt	ord_date	customer_id	salesman_id
70008	5760.00	2012-09-10	3002	5001
70013	3045.60	2012-04-25	3002	5001
70003	2480.40	2012-10-10	3009	5003
70005	2400.60	2012-07-27	3007	5001
70010	1983.43	2012-10-10	3004	5006
70007	948.50	2012-09-10	3005	5002
70009	270.65	2012-09-10	3001	5005
70012	250.45	2012-06-27	3008	5002
70001	150.50	2012-10-05	3005	5002
70004	110.50	2012-08-17	3009	5003
70011	75.29	2012-08-17	3003	5007
70002	65.26	2012-10-05	3002	5001
NULL	NULL	NULL	NULL	NULL

Result Grid | Form Editor | Field Types | Query Stats

Query 6: -ORDER BY

```
SELECT *
FROM orders
ORDER BY ord_date;
```

Output: -

19 22:55:05 SELECT * FROM orders ORDER BY ord_date LI... 12 row(s) returned 0.000 sec / 0.000 sec

ord_no	purch_amt	ord_date	customer_id	salesman_id
70013	3045.60	2012-04-25	3002	5001
70012	250.45	2012-06-27	3008	5002
70005	2400.60	2012-07-27	3007	5001
70004	110.50	2012-08-17	3009	5003
70011	75.29	2012-08-17	3003	5007
70007	948.50	2012-09-10	3005	5002
70008	5760.00	2012-09-10	3002	5001
70009	270.65	2012-09-10	3001	5005
70001	150.50	2012-10-05	3005	5002
70002	65.26	2012-10-05	3002	5001
70003	2480.40	2012-10-10	3009	5003
70010	1983.43	2012-10-10	3004	5006
NULL	NULL	NULL	NULL	NULL

5003

Query 7: -SUM()

```
SELECT SUM(purch_amt) AS total_revenue
FROM orders;
```

Output: -

20	22:57:35	SELECT SUM(purch_amt) AS total_revenue FRO...	1 row(s) returned	0.015 sec / 0.000 sec

Result Grid

	total_revenue
▶	17541.18

Query 8: -AVG()

```
SELECT AVG(purch_amt) AS average_order
FROM orders;
```

Output: -

21	22:59:14	SELECT AVG(purch_amt) AS average_order FRO...	1 row(s) returned	0.000 sec / 0.000 sec

Result Grid

	average_order
▶	1461.765000

Query 9: -MAX()

```
SELECT MAX(purch_amt) AS highest_order
FROM orders;
```

Output: -

23	23:01:55	SELECT MAX(purch_amt) AS highest_order FRO...	1 row(s) returned	0.000 sec / 0.016 sec

Result Grid

	highest_order
▶	5760.00

Query 10: - MIN()

```
SELECT MIN(purch_amt) AS lowest_order
FROM orders;
```

Output: -

24 23:04:09 SELECT MIN(purch_amt) AS lowest_order FROM... 1 row(s) returned 0.000 sec / 0.000 sec

Result Grid

	lowest_order
▶	65.26

Query 11: -COUNT()

```
SELECT COUNT(*) AS total_orders
FROM orders;
```

Output: -

25 23:06:00 SELECT COUNT(*) AS total_orders FROM orders ... 1 row(s) returned 0.031 sec / 0.000 sec

	total_orders
▶	12

Query 12: - GROUP BY

```
SELECT customer_id,
       SUM(purch_amt) AS total_purchase
FROM orders
GROUP BY customer_id;
```

Output: -

26 23:08:16 SELECT customer_id, SUM(purch_amt) AS t... 8 row(s) returned 0.000 sec / 0.000 sec

	customer_id	total_purchase
▶	3005	1099.00
	3002	8870.86
	3009	2590.90
	3007	2400.60
	3001	270.65
	3004	1983.43
	3003	75.29
	3008	250.45

Query 13: - GROUP BY + MAX()

```sql
SELECT customer_id,
       MAX(purch_amt) AS highest_purchase
FROM orders
GROUP BY customer_id;
```

Output: -

27 23:18:40 SELECT customer_id, MAX(purch_amt) AS hi... 8 row(s) returned 0.016 sec / 0.000 sec

customer_id	highest_purchase
3005	948.50
3002	5760.00
3009	2480.40
3007	2400.60
3001	270.65
3004	1983.43
3003	75.29
3008	250.45

Query 14: - GROUP BY + HAVING

```sql
SELECT customer_id,
       COUNT(*) AS total_orders
FROM orders
GROUP BY customer_id
HAVING COUNT(*) > 1;
```

Output: -

28 23:21:25 SELECT customer_id, COUNT(*) AS total_or... 3 row(s) returned 0.000 sec / 0.000 sec

customer_id	total_orders
3005	2
3002	3
3009	2

Query 15: - GROUP BY + HAVING + ORDER BY

```sql
SELECT customer_id,
       SUM(purch_amt) AS total_purchase
FROM orders
GROUP BY customer_id
HAVING SUM(purch_amt) > 1000
ORDER BY total_purchase DESC;
```

Output: -

29 23:23:45 SELECT customer_id, SUM(purch_amt) AS t... 5 row(s) returned 0.000 sec / 0.000 sec

customer_id	total_purchase
3002	8870.86
3009	2590.90
3007	2400.60
3004	1983.43
3005	1099.00

Query 16: - BETWEEN + HAVING

```sql
SELECT customer_id,
       MAX(purch_amt) AS max_purchase
FROM orders
GROUP BY customer_id
HAVING MAX(purch_amt) BETWEEN 2000 AND 6000;
```

Output: -

30 23:25:35 SELECT customer_id, MAX(purch_amt) AS m... 3 row(s) returned 0.000 sec / 0.000 sec

customer_id	max_purchase
3002	5760.00
3009	2480.40
3007	2400.60

Query 17: - MAX + GROUP BY + HAVING

```sql
SELECT ord_date,
       MAX(purch_amt) AS highest_purchase
FROM orders
GROUP BY ord_date
HAVING MAX(purch_amt) > 2000;
```

Output: -

31 23:27:25 SELECT ord_date, MAX(purch_amt) AS high... 4 row(s) returned 0.016 sec / 0.000 sec

ord_date	highest_purchase
2012-10-10	2480.40
2012-07-27	2400.60
2012-09-10	5760.00
2012-04-25	3045.60